
Multi-Agent Reinforcement Learning for Local Energy Markets: Congestion Management

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Introduction

With the growing penetration of distributed renewable resources, local energy markets (LEMs) emerge as a key paradigm for coordinating heterogeneous actors (producers, consumers, and prosumers) within constrained distribution grids. Ensuring both efficiency and reliability is challenging, as individual optimization often conflicts with global objectives, particularly under congestion.

While distributed optimization methods such as ADMM provide rigorous coordination mechanisms, they lack adaptability to uncertainty and evolving agent behavior. Multi-Agent Reinforcement Learning (MARL) offers a complementary approach, enabling agents to learn adaptive strategies that align local incentives with system-wide stability.

A central challenge lies in the role of the Distribution System Operator (DSO), whose primary lever to influence the market is dynamic pricing. Designing price signals that simultaneously mitigate congestion and activate flexibility resources represents a critical step toward resilient and fair LEM operation.

This project therefore focuses on MARL-based DSO price optimization with flexibility, extending prior work on learning-based congestion management to incorporate storage and demand response. The objective is to develop adaptive and scalable pricing strategies that ensure efficient, grid-compliant coordination in decentralized energy systems.

2 Objectives

The primary objective of this work was to investigate how **multi-agent reinforcement learning (MARL)** can be integrated into **local energy markets (LEMs)** to enhance congestion management in distribution networks. Specifically, the research sought to:

- Develop a **distributed optimization framework** based on the **Alternating Direction Method of Multipliers (ADMM)** for coordinating autonomous agents under network constraints.
- Design and implement a **learning-based control mechanism** using reinforcement learning (RL) to dynamically adjust price signals in order to mitigate grid congestion.
- Explore **reward function design** as a means to improve the robustness of the RL agent, particularly under heterogeneous and dynamic agent behavior.
- Investigate the **feasibility and effectiveness** of controlling congestion solely through DSO-to-agent price interactions, reflecting practical operational constraints.

3 Progress

The study progressed through three major stages:

1. Centralized to Distributed Market Modeling

The initial step involved formulating the LEM as an optimization problem under **convex cost functions and symmetric exchange constraints**. This problem was decomposed using **ADMM**, enabling **peer-to-peer coordination** through local computations and global synchronization. Comparative analysis confirmed the equivalence of centralized and distributed

solutions, while demonstrating improved convergence speed in the multi-agent setup due to explicit consensus enforcement.

2. Integration of Reinforcement Learning for Congestion Management

A **Soft Actor-Critic (SAC)** agent was trained to generate **dynamic price signals** influencing agent decisions. Multiple **reward function designs** were tested:

- An initial formulation penalizing constraint violations beyond the threshold.
- A **Gaussian-shaped continuous reward** centered on an optimal operating point, which yielded significant improvements in stability and congestion reduction ($\approx 23.4\%$ improvement compared to the initial reward).

3. Deployment in a Multi-Agent Environment

Despite differences in observation spaces between centralized and distributed configurations, the controller maintained effectiveness, reducing congestion violations by approximately **90% compared to a baseline without RL control**.

4 Scope and Goals

4.1 General Scope and Main Objectives

This work primarily addressed **congestion management** in **peer-to-peer energy markets** through the combination of **distributed optimization** and **learning-based control mechanisms**.

The scope was deliberately restricted to:

- A small-scale LEM comprising a producer, consumer, and DSO, with a **community manager agent** for global synchronization (Figure 1).

- Price-based coordination as the sole lever for influencing agent behavior, with price signals applied to all exchanges between agents, not just with the DSO, without any direct control over peer-to-peer transactions.
- Evaluation of **reward shaping strategies** to enhance RL policy robustness in static and dynamic environments.

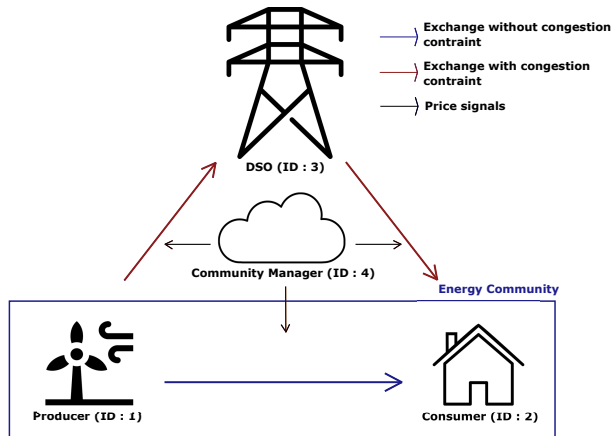


Figure 1: *Simplified Local Energy Market under Study*

The next research phase focuses on **congestion management through DSO price signals only**. Unlike the previous setup where all exchanges between agents could be influenced by pricing, the new objective is to restrict control actions to **grid exchange prices with the DSO**. Peer-to-peer transactions remain unaffected by direct intervention.

This change is motivated by the need to make the learning framework **closer to real-world market practices**, where the DSO has no authority over peer-to-peer trades and can only influence the system through tariffs and price adjustments at the grid interface. By limiting the control scope in this way, the learning agent's actions become more realistic, while still allowing effective congestion management.

The research will therefore investigate how reinforcement learning can be used to:

- Adjust DSO exchange prices dynamically to mitigate local congestion.

- Ensure cost-efficient operation for market participants without distorting peer-to-peer market autonomy.
- Improve overall system stability by shaping demand and supply profiles through price signals only at the DSO interface.

Ultimately, this will pave the way for a **multi-agent learning architecture** where:

- A DSO-controlled agent manages congestion exclusively through grid price signals.
- Other agents (producer and consumer) adapt their behaviors in response, within a decentralized market governed by fairness and efficiency principles.

4.2 Specific Achievements and Future Research Directions

Implementation Status and Current Achievements

- An initial environment has been developed, where DSO price signals influence only the exchanges with the grid, leaving peer-to-peer transactions uncontrolled but results are not satisfying.
- Preliminary simulations confirm the feasibility of using this setup to study congestion management through price adjustments at the grid interface.
- The current framework is flexible and can be extended to test different operating conditions and market configurations.

Future Objectives

- Develop reinforcement learning strategies to reduce the power exchanged with the DSO by influencing agent behavior solely through dynamic price signals applied at the DSO interface.

- Identify and calibrate appropriate ranges of DSO price signals, ensuring they are strong enough to influence agent decisions while still allowing system convergence.
- Design and evaluate alternative reward functions, selecting the most suitable one through a systematic and methodological process to enhance learning efficiency and stability.
- Study the learning process itself with greater precision, carefully selecting the most relevant training parameters to ensure robustness, faster convergence, and meaningful policy outcomes.
- Explore and compare other potential control or learning methods that could outperform reinforcement learning in managing congestion through DSO price-based coordination.